Check-In 01/14. *(True/False)* The vector $\mathbf{u} = \langle 2, -1, 0 \rangle$ is a unit vector.

Solution. The statement is *false*. We know a unit vector has length 1. We know if $\mathbf{v} = \langle x_1, x_2, \ldots, x_n \rangle$, then $\|\mathbf{v}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$. But then $\|\mathbf{u}\| = \|\langle 2, -1, 0 \rangle = \sqrt{2^2 + (-1)^2 + 0^2} = \sqrt{4+1+0} = \sqrt{5} \neq 1$. Therefore, $\mathbf{u}$ is not a unit vector.

Check-In 01/16. *(True/False)* Suppose that $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. If $\mathbf{u} \cdot \mathbf{v} = 0$, then either $\mathbf{u} = 0$ or $\mathbf{v} = 0$. Furthermore, $\mathbf{u} \perp \mathbf{v}$.

Solution. The statement is *false*. If $\mathbf{u} \cdot \mathbf{v} = 0$, it is true that $\mathbf{u}$ and $\mathbf{v}$ are perpendicular (so long as neither of them are nonzero). Furthermore, if $\mathbf{u}$ and $\mathbf{v}$ are perpendicular, then $\mathbf{u} \cdot \mathbf{v} = 0$. However, if $\mathbf{u} \cdot \mathbf{v} = 0$, it need not be the case that $\mathbf{u}$ and $\mathbf{v}$ are zero. For instance, if $\mathbf{u} = \langle 1, 0, \ldots, 0 \rangle$ and $\mathbf{v} = \langle 0, 1, 0, \ldots, 0 \rangle$, then $\mathbf{u} \cdot \mathbf{v} = 1(0) + 0(1) + 0(0) + \cdots + 0(0) = 0$ but neither $\mathbf{u}$ nor $\mathbf{v}$ are zero. Furthermore, it is impossible that $\mathbf{u} = 0$ or $\mathbf{v} = 0$. Both $\mathbf{u}, \mathbf{v}$ are *vectors* while 0 is a scalar.

Check-In 01/21. *(True/False)* The augmented matrices $\begin{pmatrix} 1 & 2 & 3 & 0 \\ -1 & 5 & 1 & 5 \\ 1 & -1 & 1 & -2 \end{pmatrix}$ and $\begin{pmatrix} -1 & 5 & 1 & 5 \\ 1 & 2 & 3 & 0 \\ 2 & -2 & 2 & -4 \end{pmatrix}$ represent equivalent systems of linear equations.

Solution. The statement is *true*. We know that for systems of linear equations, we can represent these systems as augmented matrices. Elementary row operations on these matrices correspond to viable arithmetic operations on the system of equation 'side.' These operations preserve the solutions (if any) to the system of equations, i.e. produce equivalent systems. So, we can perform any elementary row operation on these augmented matrices and obtain equivalent systems: interchange rows, scale a row by a nonzero scalar, and replace any single row with a linear combination of other rows. Observe we can obtain the second matrix from the first by interchanging the first two rows and scaling the third row by 2. Therefore, the systems of linear equations represented by these matrices are equivalent. If the first matrix was an augmented matrix that was 'untouched' from the original system. The system of equations was...

$$\begin{cases} x + 2y + 3z = 0 \\ -x + 5y + z = 5 \\ x - y + z = -2 \end{cases}$$

which has solution $(-\frac{1}{2}, 1, -\frac{1}{2})$.

Check-In 01/23. *(True/False)* There exists a linear system consisting of some number of equations with some number of variables such that there are five solutions to the given system.

Solution. The statement is *false*. For a system of linear equations, there are only three possibilities: there are no solutions to the system of equations, there is *one* solution to the system of equations, or there are infinitely many solutions. It is not possible to have any other number of solutions.

Check-In 01/26. *(True/False)* If a system of linear equations has a unique solution, then the RREF of the augmented matrix has a pivot position in every column except for the last one—which corresponds to the solution.

Solution. The statement is *true*. Denote the coefficient matrix by A. If the augmented matrix has a pivot position in every column except for the last one, then A has a pivot position in every column. We know each column corresponds to variable in the original system of equations. We also know that each pivot column corresponds to a basic variable, i.e. one whose value is either fixed or given in terms of a free variable. But because every column has a pivot position, we know each of the variables in the system are basic variables, i.e. there are no free variables. Therefore, each of the variables has a fixed value, i.e. there is a unique solution.

Check-In 01/28. *(True/False)* Consider a matrix equation $A\mathbf{x} = \mathbf{b}$. If the RREF of A has no zero rows, then the matrix-vector equation has a solution.

Solution. The statement is *true*. If the RREF of A has no zero rows, there can be no row in the RREF augmented matrix $[A\ \ \mathbf{b}]$ can have no row of the form $[0\ \ 0\ \ \cdots\ \ 0\ \ 1]$. But then the original system cannot be inconsistent, i.e. it is consistent. Therefore, there must be at least one (possibly infinitely many) solutions to the original system of equations.

Check-In 01/30. *(True/False)* If REF of an augmented matrix coming from a system of linear equations is $\begin{pmatrix} 1 & 1 & 1 & -1 & 0 \\ 0 & 1 & -2 & 1 & 4 \\ 0 & 0 & 0 & 2 & 6 \end{pmatrix}$, then the solution is $(x_1, x_2, x_3, x_4) = (2 - 3t, 2t + 1, t, 3)$, where t is any real number.

Solution. The statement is *true*. From the last row, we know that $2x_4 = 6$, i.e. $x_4 = 3$. From the second row, we know that $x_2 - 2x_3 + x_4 = 4$. But we know $x_4 = 3$. Therefore, $x_2 - 2x_3 + 3 = 4$, which implies $x_2 = 2x_3 + 1$. Finally, from the first row, we know $x_1 + x_2 + x_3 - x_4 = 0$. Using the fact that $x_4 = 3$ and $x_2 = 2x_3 + 1$, we know $x_1 + (2x_3 + 1) + x_3 - 3 = 0$, which implies $x_1 = 2 - 3x_3$. Letting $x_3 = t$, we have $(x_1, x_2, x_3, x_4) = (2 - 3t, 2t + 1, t, 3)$.

Check-In 02/04. *(True/False)* If a 10×10 matrix A has rank 7, then it cannot be invertible. Furthermore, there must be a vector $\mathbf{b} \in \mathbb{R}^{10}$ such that $A\mathbf{x} = \mathbf{b}$ has no solution.

Solution. The statement is *true*. We know that A has an inverse if there is a matrix B such that $AB = I$, where I is the identity matrix. This occurs if and only if we can form an augmented matrix $[A\ \ I]$ and place A into RREF and obtain an identity matrix. If so, then A^{-1} appears in the columns where I was in this augmented matrix. If A has rank 7, then has only 7 pivot positions in its RREF. But then the RREF of A cannot be the identity matrix, which in this case would have 10 pivot positions. Therefore, A is not invertible. Moreover, because A has only 7 pivot positions in its RREF, there must be at least one row of zeros in its RREF. But then we can find a $\mathbf{b}$ vector such that, after these row reductions, we have a row in the RREF of the augmented matrix $[A \;\; \mathbf{b}]$ of the form $[0 \;\; 0 \;\; \cdots \;\; 0 \;\; 1]$. But then the system $A\mathbf{x} = \mathbf{b}$ has no solution.

Check-In 02/06. *(True/False)* $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$

Solution. The statement is *true*. Indeed, this is how we define the determinant for a two-by-two matrix. We then define the determinant of larger $n \times n$ matrices via cofactor expansions, which eventually reduces the problem of finding the given determinant into computing determinants of various 2×2 matrices.

Check-In 02/09. *(True/False)* If A, B are $n \times n$ matrices, B is invertible, and $\det(AB) = 0$, then it *must be* that A is not invertible.

Solution. The statement is *true*. We know that a square matrix A is invertible if and only if $\det A \neq 0$. We also know that $\det(AB) = \det A \cdot \det B$. Because $0 = \det(AB) = \det A \cdot \det B$, either $\det A = 0$ or $\det B = 0$. Because B is invertible, we know $\det B \neq 0$. But then it must be that $\det A = 0$, which implies that A is not invertible.

Check-In 02/18. *(True/False)* An LU factorization for a matrix A is a decomposition of A into a product of the form $A = LU$, where L is lower triangular and U is upper triangular.

Solution. The statement is *true*. This is the definition of an LU factorization/decomposition. In fact, one typically also requires the matrix L to be unit lower triangular, i.e. the diagonal entries of L to be equal to 1. One of the 'standard' approaches (the one we introduced in class) to computing L ensures this.

Check-In 02/20. *(True/False) Every* $n \times m$ matrix A has a PLU factorization.

Solution. The statement is *true*. Computing an LU decomposition is essentially computing the row echelon form of a matrix without row interchanges (or often scaling). Of course, this is not always possible, so not all matrices have an LU factorization. But if one allows row interchanges, one can always find the REF of a matrix. But then one should always be able to find a PLU factorization for a matrix. A square, invertible matrix has an LU decomposition if and only if all the leading principal minors are nonzero. More generally, if a square matrix is singular, i.e. not invertible, and has rank k, then it will have an LU decomposition if the first k leading principal minors are nonzero—although the reverse need not be true. Lastly, we know a if square, invertible matrix A has an LU decomposition (with 1s along the diagonal of L), then this decomposition is unique.

Check-In 02/23. *(True/False)* If $P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$, then PA interchanges performs the operations $R_1 \leftrightarrow R_4$ and $R_2 \leftrightarrow R_3$.

Solution. The statement is *false*. We can interpret left multiplication by a square matrix A, i.e. AB, as 'acting' on the rows of B. This is how we interpret left multiplication of B by an elementary/permutation matrix, EB or PB, respectively. This is also how we can easily construct elementary/permutation matrices with a desired effect. We can label the columns and rows of B with labels R_i. The row labels will represent the ith row of the result EB or PB. The entries in that row indicate the linear combination of the original rows of B from which the ith row of the product is formed. For instance, if the 3rd row of E in EB is $(0 \ \ -3 \ \ 5 \ \ 7)$, then the 3rd row of the product will be $-3R_2 + 5R_3 + 7R_4$, where R_i is the ith row of B. Examining the given P in this way, we have...

$$P = \begin{array}{c} \\ \widetilde{R_1} \\ \widetilde{R_2} \\ \widetilde{R_3} \\ \widetilde{R_4} \end{array} \!\!\begin{array}{c} \begin{array}{cccc} R_1 & R_2 & R_3 & R_4 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

So, the new Row 1 will be the same as the original Row 1, the new Row 2 will be the original Row 3, the new Row 3 will be the original Row 2, and the new Row 4 will be the original Row 4. Therefore, this matrix interchanges rows two and three of the original matrix, i.e. P represents the row operation $R_2 \leftrightarrow R_3$. The matrix which would perform $R_1 \leftrightarrow R_4$ and $R_2 \leftrightarrow R_3$ would be...

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

Check-In 02/25. *(True/False)* The collection of vectors $\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$ is linear independent.

Solution. The statement is *false*. Foremost, *any* collection of vectors containing the zero vector, $\mathbf{0}$, is *never* linearly independent. We know that $c \cdot \mathbf{0} = \mathbf{0}$ for any real number c. Furthermore, suppose $\{\mathbf{0}, v_1, \ldots, v_n\}$ is a collection of vectors. Then $1 \cdot \mathbf{0} + 0v_1 + 0v_2 + \cdots + 0v_n = \mathbf{0}$ and $0 \cdot \mathbf{0} + 0v_1 + 0v_2 + \cdots + 0v_n = \mathbf{0}$ are different representations of $\mathbf{0}$ using the vectors from the collection. So, the collection would be linearly dependent. Alternatively, we see that the second vector is a scalar multiple of the first, i.e. parallel to the first. A collection of vectors in which two or more vectors are parallel can *never* be linearly independent. Suppose $\{v_1, \ldots, v_n\}$ is a collection of vectors where (after possibly renumbering them) v_1 and v_2 are parallel. But then $v_1 = cv_2$. But then $0v_1 + 0v_2 + \cdots + 0v_n = \mathbf{0}$ and $-1v_1 + cv_2 + 0v_3 + \cdots + 0v_n = -cv_2 + cv_2 + 0v_3 + \cdots + 0v_n = \mathbf{0}$ are different representations of $\mathbf{0}$ using the vectors in $\{v_1, \ldots, v_n\}$. Therefore, the collection of vectors is linearly dependent. In our specific case, we can see that $-4 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 4 \\ 4 \end{pmatrix} = \mathbf{0}$, $\begin{pmatrix} 1 \\ 1 \end{pmatrix} - \frac{1}{4} \begin{pmatrix} 4 \\ 4 \end{pmatrix} = \mathbf{0}$, and

$0\begin{pmatrix}1\\1\end{pmatrix}+0\begin{pmatrix}4\\4\end{pmatrix}=\mathbf{0}$ are all distinct representations of $\mathbf{0}$. Therefore, the collection $\left\{\begin{pmatrix}1\\1\end{pmatrix},\begin{pmatrix}4\\4\end{pmatrix},\begin{pmatrix}0\\0\end{pmatrix}\right\}$ cannot be linearly independent, i.e. the collection is linearly dependent.

Check-In 02/27. *(True/False)* A basis for a subspace is a collection of vectors that span the subspace.

Solution. The statement is *false*. For a collection of vectors to be a basis for a subspace, the collection needs to span the subspace and be linearly independent. The collection $\left\{\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}0\\1\end{pmatrix},\begin{pmatrix}1\\1\end{pmatrix}\right\}$ spans $\mathbb{R}^2$. However, this collection is *not* a basis for $\mathbb{R}^2$ because the set is not linearly independent because $1\begin{pmatrix}1\\0\end{pmatrix}+1\begin{pmatrix}0\\1\end{pmatrix}+(-1)\begin{pmatrix}1\\1\end{pmatrix}=\mathbf{0}$ is a nontrivial way, i.e. not all weights 0, of obtaining $\mathbf{0}$. Therefore, the collection of vectors is linearly dependent. A basis needs to be a linearly independent spanning set for a subspace. A collection of vectors spanning a subspace means that every vector in the subspace can be obtained using a linear combination of vectors from the collection. The set being linearly independent means that the linear combination of vectors from the collection giving the desired vector is unique, i.e. there is only one way to obtain a vector in the subspace 'using' vectors from the collection. So, a basis for a subspace is a collection of vectors that allows one to obtain each vector in the subspace uniquely using a linear combination of those vectors. As we shall see, this also means the basis can serve as a 'coordinate system' for the subspace.

Check-In 03/02. *(True/False)* The nullspace of a matrix forms a subspace of some vector space.

Solution. The statement is *true*. Recall that the nullspace of a $m\times n$ matrix is the set of solutions to the homogeneous equation $A\mathbf{x}=\mathbf{0}$. That is, $\operatorname{null} A=\{\mathbf{x}\in\mathbb{R}^n\colon A\mathbf{x}=\mathbf{0}\}$. Recall that S is a subspace of a vector space V if...

- $\mathbf{0}\in S$ (the zero vector is in the subspace): $\mathbf{0}\in\operatorname{null} A$ because $A\mathbf{0}=\mathbf{0}$.
- if $u,v\in S$, then $u+v\in S$ (you can add vectors in the subspace): Suppose $\mathbf{x},\mathbf{y}\in\operatorname{null} A$. So, $A\mathbf{x}=\mathbf{0}$ and $A\mathbf{y}=\mathbf{0}$. But then $A(\mathbf{x}+\mathbf{y})=A\mathbf{x}+A\mathbf{y}=\mathbf{0}+\mathbf{0}=\mathbf{0}$, so that $\mathbf{x}+\mathbf{y}\in\operatorname{null} A$.
- if $c\in\mathbb{R}$ and $v\in S$, then $cv\in S$ (you can scale vectors in the subspace): Let $c\in\mathbf{R}$ and $\mathbf{x}\in\operatorname{null} A$. Because $\mathbf{x}\in\operatorname{null} A$, $A\mathbf{x}=\mathbf{0}$. But then $A(c\mathbf{x})=c(A\mathbf{x})=c\mathbf{0}=\mathbf{0}$, so that $c\mathbf{x}\in\operatorname{null} A$.

Therefore, the nullspace of a $m\times n$ matrix A is a subspace of $\mathbb{R}^n$.

Check-In 03/04. *(True/False)* If the nullspace of A has dimension 2, then the equation $A\mathbf{x}=\mathbf{b}$ must have infinitely many solutions.

Solution. The statement is *false*. Consider the following example with $A=\begin{pmatrix}1&0&0\end{pmatrix}$ and consider the equation $A\mathbf{x}=\begin{pmatrix}0\\0\\1\end{pmatrix}$. We can see that A has nullity 2, so the dimension of its nullspace is also 2.

However, the equation has no solutions. Indeed, if $\mathbf{x} = \begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix}^T$ were a solution, then...

$$\begin{gathered} A\mathbf{x} = \mathbf{b} \\ \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\ \begin{pmatrix} x_1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \end{gathered}$$

But this implies $0 = 1$, which is impossible. So, there is no solution. Recall for $A\mathbf{x} = \mathbf{b}$ to have a solution, $\mathbf{b}$ has to be in the column space of A. If that is the case, then there will be a unique solution if the dimension of the nullspace of A (the nullity of A) is 0. Otherwise, there will be infinitely many solutions. So, as long as there is a solution, once there is 'some nullity', there will be infinitely many solutions. But there has to be a solution first!

Check-In 03/06. *(True/False)* The column space of A is *always* generated by the span of the columns of A but may not be a basis for the column space.

Solution. The statement is *true*. The first part is the definition of the column space—the column space is the span of the columns of A. But the columns of A may not be a basis for the column space. For instance, consider $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$. Clearly, the third column of A is redundant as it is a copy of the second column. So, $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ cannot be a basis for the column space. Eliminating this third column, we have $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$, which is a basis because it clearly still spans the column space (we only eliminated a redundant vector) and it linearly independent (there only two vectors and neither is a scalar multiple of the other). Notice these are the pivot columns of A. Indeed, one can always use the pivot columns of A—be careful its the pivot columns of A and not its RREF—to form a basis for the column space of A.

Check-In 03/18. *(True/False)* If A is a 3×7 matrix with rank 5, then $A\mathbf{x} = \mathbf{b}$ has a solution for all $\mathbf{x} \in \mathbb{R}^3$. However, the solution will only be unique for some $\mathbf{b}$.

Solution. The statement is *false*. First of all, if A is a $m \times n$ matrix, we know that $\operatorname{rank} A \leq \min\{m, n\}$; that is, the rank of A is at most its number of rows or number of columns—whichever is smaller. So, if A is 3×7, we know $\operatorname{rank} A \leq \min\{3, 7\} = 3$, which means $\operatorname{rank} A \neq 5$. Furthermore, if A is a $m \times n$ matrix, by the Rank-Nullity Theorem, we know that $\operatorname{rank} A + \operatorname{nullity} A = n$. But then whether $\operatorname{rank} A = 5$ or $\operatorname{rank} A \leq 3$, we know that $\operatorname{nullity} A > 0$. So, there is 'some nullity.' Therefore, whenever $A\mathbf{x} = \mathbf{b}$ has a solution, we know it will have infinitely many solutions! Finally, because $\operatorname{rank} A = 5$ is impossible, we would need to know what the rank actually is to know whether or not $A\mathbf{x} = \mathbf{b}$ has a solution. If $\operatorname{rank} A = 3$, we know that $A\mathbf{x} = \mathbf{b}$ will have a solution for all $\mathbf{b} \in \mathbb{R}^3$. However, if $\operatorname{rank} A < 3$, there will only be a solution to $A\mathbf{x} = \mathbf{b}$ for some $\mathbf{b} \in \mathbb{R}^3$.

Check-In 03/20. *(True/False)* A collection of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly dependent if and only if the matrix formed by the vectors, $A = [\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_n]$, has positive nullity.

Solution. The statement is *true*. We know that a collection of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly dependent if there is a nontrivial solution to $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$. If we form a matrix A using the $\mathbf{v}_i$ as the columns, i.e. $A = [\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_n]$, then this occurs if and only if $A\mathbf{c} = \mathbf{0}$ has a nontrivial solution, where $\mathbf{c} = \begin{pmatrix} c_1 & c_2 & \cdots & c_n \end{pmatrix}^T$. The equation $A\mathbf{c} = \mathbf{0}$ always has the trivial solution $\mathbf{0}$. So, if there is any other solution, there will be infinitely many solutions. But then $\text{nullity}\, A > 0$. But then $A\mathbf{c} = \mathbf{0}$ has a nontrivial solution if and only if $\text{nullity}\, A > 0$, i.e. the nullity of A is positive.

Check-In 03/23. *(True/False)* If A is a $n \times n$ invertible matrix, then $\dim\ \text{Col}\ A = n$.

Solution. The statement is *true*. Think of how we can determine whether a $n \times n$ matrix is invertible. We augment A with the identity matrix, I_n, and place the matrix 'on the left' (the original A) into RREF. We know that A is invertible if and only if we obtain the identity matrix 'on the left.' If so, the inverse, A^{-1}, is the matrix 'on the right.' But the rank of the identity matrix is n. Row reduction does not change the rank of the resulting matrix and a matrix's rank is the dimension of the column space. So, we know a $n \times n$ matrix A is invertible if and only if $n = \text{rank}\, A = \dim\ \text{Col}\ A$.

Check-In 03/25. *(True/False)* The matrix associated to the linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ given by $T(x, y) = (x + y, y - x)$ is $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.

Solution. The statement is *false*. Write the transformation as vectors rather than the typical 'input-output form':

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ y - x \end{pmatrix}$$

Although the 'matrix' on the right is only a vector—which has only a single column, we pretend it is a matrix with several columns—corresponding to the variables in a given order. So, aligning the 'columns' of the output, we have...

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x & + & y \\ -x & + & y \end{pmatrix}$$

The coefficients of the variables in this resulting 'matrix' is exactly the matrix associated to the linear transformation. Therefore, the matrix associated to our given T is...

$$A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

The given result did not properly align the variables.

Check-In 04/01. *(True/False)* An eigenvector for a square matrix A is *any* vector $\mathbf{v}$ such that $A\mathbf{v} = \lambda\mathbf{v}$, where λ is a scalar called the eigenvalue associated to $\mathbf{v}$.

Solution. The statement is *false*. The zero vector satisfies the statement because $A\mathbf{0} = \lambda\mathbf{0} = \mathbf{0}$ for any choice of λ. However, the definition of an eigenvector precludes the zero vector from being an eigenvector, "An eigenvector is a *nonzer* vector $\mathbf{v}$ such that $A\mathbf{v} = \lambda\mathbf{v}$, where λ is some scalar called the eigenvalue associated to $\mathbf{v}$." Therefore, $\mathbf{v}$ cannot be *any* vector—it must be nonzero. [This is important for the uniqueness of λ and for some of our uses of eigenvectors.]

Check-In 04/03. *(True/False)* An eigenvalue associated to an eigenvector is unique. However, there are many eigenvectors associated to a fixed eigenvalue.

Solution. The statement is *true*. We have seen that an eigenvalue associated to an eigenvector is unique. If an eigenvector $\mathbf{v}$ were such that $A\mathbf{v} = \lambda_1\mathbf{v}$ and $A\mathbf{v} = \lambda_2\mathbf{v}$, then because both are equal to $A\mathbf{v}$, we know $\lambda\mathbf{v} = \lambda_2\mathbf{v}$. This implies $(\lambda_1 - \lambda_2)\mathbf{v} = \mathbf{0}$. Because $\mathbf{v}$ is not the zero vector (it is an eigenvector and cannot be $\mathbf{0}$ by definition), it must be $\lambda_1 - \lambda_2 = 0$, i.e. $\lambda_1 = \lambda_2$, i.e. the eigenvalue associated to $\mathbf{v}$ is unique. However, an eigenvector associated to λ is merely a nonzero vector such that $A\mathbf{v} = \lambda\mathbf{v}$, i.e. $(A - \lambda I_n)\mathbf{v} = \mathbf{0}$. That is, $\mathbf{v} \in (A - \lambda I_n)$. Because we know this nullspace must consist of more than just the zero vector, there are infinitely many choices for $\mathbf{v}$—any nonzero element of the nullspace will do! For instance, consider the example, let $A = \begin{pmatrix} -4 & -3 \\ 3 & 6 \end{pmatrix}$. This matrix has eigenvalues $\lambda = -3, 5$. The eigenspace associated to $\lambda = -3$ has basis $\left\{ \begin{pmatrix} -3 \\ 1 \end{pmatrix} \right\}$. Therefore, any nonzero vector in the span of this set can serve as an eigenvector for $\lambda = -3$. For instance, $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$, $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$, $\begin{pmatrix} -6 \\ 2 \end{pmatrix}$, etc.

Check-In 04/06. *(True/False)* If the characteristic polynomial of a matrix A is $p(\lambda) = (\lambda+3)^2(\lambda-1)$, then $\lambda = -3$ is an eigenvalue with multiplicity 2 and its eigenspace has dimension two.

Solution. The statement is *false*. We can see that the eigenvalues of A are $\lambda = -3$ and $\lambda = 1$, i.e. the solutions to $p(\lambda) = 0$. Because the factor of $p(\lambda)$ for $\lambda = -3$ has degree 2, we know that the (algebraic) multiplicity of $\lambda = -3$ is 2. However, without computing the eigenspace, we cannot say that its dimension is 2. The dimension of the eigenspace for λ is its geometric multiplicity. All we know is that the geometric multiplicity of λ is *at most* its algebraic multiplicity (and is at least 1, so here it is either 1 or 2). But without knowing A, we cannot say for sure that it has dimension 2. In fact, take $A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. The matrix is diagonal, so its eigenvalues are its diagonal entries—1 and -3. One can easily verify through routine calculation that the dimension of the eigenspace for $\lambda = 3$ is 1—despite it having algebraic multiplicity 2.

Check-In 04/08. *(True/False)* If A is a 6×6 matrix with eigenspaces of dimension $1, 2, 3$, respectively, then A is diagonalizable.

Solution. The statement is *true*. We know that an $n \times n$ matrix is diagonalizable if and only if it has n linearly independent vectors. This will occur if and only if the algebraic multiplicity of each of its eigenvalues matches their geometric multiplicity. The geometric multiplicity of an eigenvector is the dimension of its eigenspace. So, one can find $1 + 2 + 3 = 6$ linearly independent vectors for A. Therefore, A is diagonalizable.